Př

$f(x)$	$f'(x)$	$\nabla f(x)$
$\|x\|$	$x^T / \|x\|$	$\nabla f(x) = (f')^T$
$\|x\|^2$	$2x^T$	$(f'(x))^T$
$x^T A x$	$x^T(A+A^T)$ | $2x^T A$	— " —
$g(x)^T g(x)$	$2g(x)^T g'(x)$	— " —
$g(x)^T h(x)$	! $h^T(x) g'(x) + g(x)^T h'(x)$	— " —
$g(x)^T A g(x)$	$g(x)^T(A+A^T) g'(x)$ $2g(x)^T A g'(x)$	— " —

A je symetrická

Př $\min x^T A x$, A je symetrická pos. def
zp. $b^T x = 1$

$L(x,\lambda) = x^T A x - \lambda(b^T x - 1)$

$\frac{\partial L}{\partial x} = 2x^T A - \lambda b^T$

$\frac{\partial L}{\partial \lambda} = -b^T x + 1$

$\frac{\partial L}{\partial x} = 0 \to x^T A = \frac{\lambda b^T}{2} \to Ax = \frac{b\lambda^T}{2} \to x = A^{-1}\frac{b\lambda^T}{2}$

$\frac{\partial L}{\partial \lambda} = 0 \to b^T A^{-1} b \lambda^T = 2 \to \lambda$ je skalár

$\lambda = \frac{2}{b^T A^{-1} b}$

$$\boxed{x = \frac{A^{-1} b}{b^T A^{-1} b}}$$

Př Taylor. poly. 0, 1, 2 stupně pro $f(x,y) = x^{-1} + y^{-1} + xy$, v bodě $(x_0, y_0) = (2, 1)$

T^0

$f(x_0, y_0) = \frac{1}{2} + 1 + 2 = \frac{7}{2}$

T^1

$f(x_0,y_0) + f'(x_0, y_0)\begin{pmatrix} x - x_0 \\ y - y_0 \end{pmatrix} = \frac{7}{2} + \begin{pmatrix} -\frac{1}{4} + 1 = \frac{3}{4} \\ -1 + 2 = 1 \end{pmatrix}^T \begin{pmatrix} x-2 \\ y-1 \end{pmatrix} = \frac{7}{2} + \frac{3}{4}(x-2) + (y-1)$

T^2 $f(x_0,y_0) + f'(x_0, y_0)\begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix} + \frac{1}{2}\begin{bmatrix} x - x_0 & y - y_0 \end{bmatrix} f''(x_0, y_0) \begin{bmatrix} x - x_0 \\ y - y_0 \end{bmatrix}$

Př

$$A = \frac{1}{15}\begin{bmatrix} -13 & 2 & -22 \\ -16 & 14 & -4 \end{bmatrix}, \text{ najděte } B, \operatorname{rank}(B) = 1$$

a) $AA^T = V \Lambda V^T$; $V = \begin{bmatrix} 0{,}6 & -0{,}8 \\ -0{,}8 & -0{,}6 \end{bmatrix}$ $\Lambda = \begin{bmatrix} 1 & 0 \\ 0 & 4 \end{bmatrix}$

Y max Λ

$\boxed{B = YY^T A}$ $\to \min \|A - B\| = \sqrt{\min \Lambda}$

b) $A = USV^T$; $V = \begin{bmatrix} \boxed{\begin{matrix} 0{,}8 \\ 0{,}6 \end{matrix}} & \boxed{\begin{matrix} 0{,}6 \\ -0{,}8 \end{matrix}} \end{bmatrix}$; $S = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$, $V = [\ \cdots\ ]$

$B = U \cdot \begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \end{bmatrix}$